Where a Mind-Migration Pipeline Breaks: A Functional-Preservation Audit of Partition, Transport, and Local-Rule Writing

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Abstract

Speculative "mind-migration" pipelines propose partitioning a brain into a regenerable Base-OS and a transferable identity payload, compressing the payload, aligning it to a new substrate, and writing it in with local plasticity. We ask whether each stage is coherent in simulation, using a Functional Preservation Score (FPS) — a null-normalised measure of how much input–output function survives an operation — on the *C. elegans* connectome and small recurrent networks that run on a laptop. All three migration stages fail, for largely independent reasons: the identity payload is not a functionally separable module ($p \geq 0.16$); Gromov–Wasserstein transport collapses once the target's real wiring is respected (fidelity $1.00 \to 0.40$, $p < 10^{-3}$) or alignment is imperfect (0.06); and biologically plausible local plasticity cannot reach the function the target topology provably admits (gap 0.45, $p < 10^{-3}$, $d = 2.0$), with sleep-ripple timing providing no benefit. The common obstacle is the target substrate's own wiring. We frame this as a reusable recipe — make preservation measurable, verify the metric has headroom over a trivial null, and test each pipeline stage against an achievable upper bound — for failure analysis of ambitious biological-ML pipelines.

1 Motivation

Ambitious biological-ML pipelines are often argued as compositions of individually plausible steps. Mind migration is an extreme case: each step borrows a real result, but the composition is untested and several steps have no evidence at any scale. We take the computational core and ask a single measurable question per stage — does it preserve function? — and report where it breaks. This short paper focuses on the negative result (the three-stage pipeline); the companion work also characterises connectome and replay compression, which succeed.

2 Method: FPS and the audit protocol

For a reference system with outputs Y_{ref} and a transformed system with outputs Y, let $\text{raw}(Y) = 1 - \|Y - Y_{\text{ref}}\|^2 / \|Y_{\text{ref}} - \overline{Y_{\text{ref}}}\|^2$, and normalise by the best trivial baseline Y_{null} (shuffle or predict-the-mean): $\text{FPS} = \text{clip}((\text{raw}(Y) - \text{raw}(Y_{\text{null}}))/(1 - \text{raw}(Y_{\text{null}})), 0, 1)$. The protocol has three rules learned the hard way: validate on real data; verify the metric beats its trivial null *before* trusting a comparison; and compare each stage to an achievable upper bound, not to perfection. Numbers are reported with bootstrap 95% CIs and permutation p-values.

3 Result: the pipeline breaks at every stage

Partition is not separable. Holding the sensory/motor interface fixed, ablating the asserted interneuron "payload" of the real *C. elegans* connectome is no more disruptive than ablating a random interneuron subset of the same size ($p = 0.17, 0.16, 0.27, 0.37$ across payload fractions 0.1–0.4; 500-sample null). Function is distributed; there is no module to lift out.

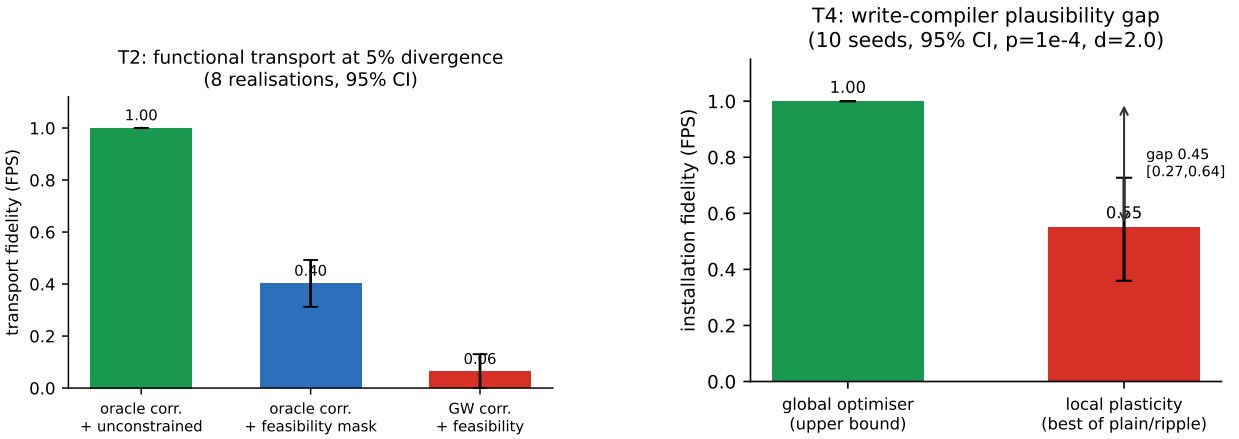


Figure 1: Left: transport fidelity at 5% divergence collapses under the feasibility mask and under estimated alignment (95% CI). Right: local plasticity cannot close the gap to the global ceiling the topology admits.

Transport breaks under realism (Figure 1, left). Entropic Gromov–Wasserstein recovers 0.76 of a ground-truth correspondence at 5% divergence, but transporting weights to a second individual preserves function only in a doubly idealised limit. With an oracle correspondence and unconstrained writing, fidelity is 1.00; once the target's wiring is respected it falls to 0.40 (95% CI $[0.31, 0.49]$, $p = 3 \times 10^{-4}$), and with the estimated correspondence to 0.06 ($p = 2 \times 10^{-4}$).

Local-rule writing cannot reach the target (Figure 1, right). A global optimiser installs the target function on the fixed topology (ceiling 1.00), but strictly local reward-modulated plasticity reaches only 0.55 ($[0.36, 0.73]$), a plausibility gap of 0.45 ($[0.27, 0.64]$, $p = 10^{-4}$, $d = 2.0$). Sleep-ripple-timed gating confers no advantage.

The three failures are independent and compound, and the target substrate's own wiring is the binding limit shared by transport and writing — the constraint the original proposal assumes away.

4 Takeaway

The pipeline is not merely unbuilt; its middle stages are incoherent in simulation on the nearest testable substrate, each for a different reason. More useful than the specific negative is the recipe: a measurable preservation score, a trivial-null headroom check, and an achievable upper bound per stage turn an unfalsifiable roadmap into a set of falsifiable claims. Scope: nematode and small-network scale; nothing here concerns reading or writing a living brain.

References